

STATSIG lecture 10

System Identification in terms of Frequency Response Functions

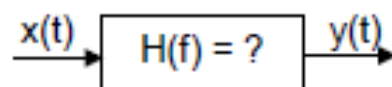
Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Non-parametric SYSTEM identification - Intro

- What:**
- Lecture #8: Spectral estimation of SIGNAL (non-parametric)
 - Lecture #9: Spectral estimation of SIGNAL (parametric)
 - where signal $x(t)$ is assumed to be measured without measurement noise

Lecture #10: Spectral estimation of SYSTEM (system identification)



- where $x(t)$ and/or $y(t)$ can only be measured including measurement noise
- non-parametric estimation of $H(f)$
- in this context often called FRF, Frequency Response Function
- non-parametric estimation often first step, e.g. recognize peaks etc.)

Lecture #11 + #12: Adaptive systems (including adaptive system identification)

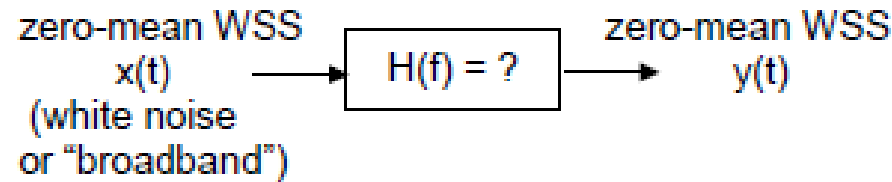
- Why:**
- Many applications in communication, acoustics, control, vibration etc., e.g.
 - estimation of transmission channel's linear distortion
 - estimation of electrical to acoustic transfer function (or vice versa)
 - estimation of control system "plant"

How: Classical methods

Frequency sweep: $x(t) = \cos(\omega t)$, $\omega \in [\omega_a, \omega_b]$
 $y(t) = |H(\omega)| \cos(\omega t + \angle H(\omega))$
sweep rate must be low enough to ensure all transients
have died out at each frequency

Impulse response: $x(t) = \delta(t) \Rightarrow y(t) = h(t)$
very little energy in output

Set-up for ideal stochastic method



$x(t)$ and $y(t)$ can be measured without measurement noise

Basic I/O relations (lecture #3 review):

$$S_{xy}(f) = H(f)S_{xx}(f)$$

$$S_{yx}(f) = S_{xy}^*(f) = H^*(f)S_{xx}(f)$$

$$S_{yy}(f) = |H(f)|^2 S_{xx}(f) = H^*(f)S_{xy}(f) = H(f)S_{yx}(f)$$

Notation

In this context is often used one-sided PSD's

$$G_{xx}(f) = 2 S_{xx}(f) u(f) = \begin{cases} 2S_{xx}(f) & f \geq 0 \\ 0 & f < 0 \end{cases}$$

likewise for $G_{yy}(f)$ and $G_{xy}(f)$

FRF estimators

$$\hat{H}_1(f) = \frac{\hat{G}_{xy}(f)}{\hat{G}_{xx}(f)} \quad \text{and}$$

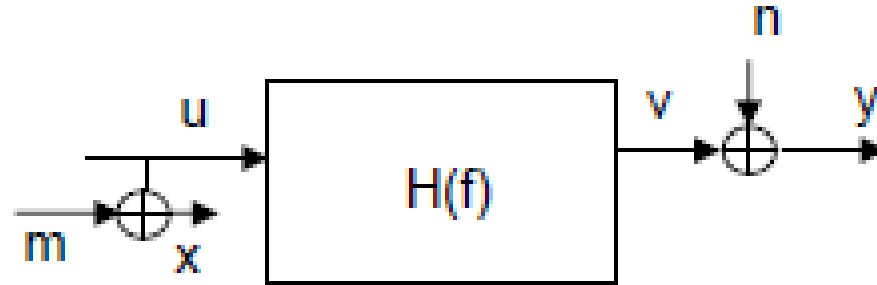
$$\hat{H}_2(f) = \frac{\hat{G}_{yy}(f)}{\hat{G}_{xy}^*(f)}$$

coherence function

$$\gamma_{xy}^2(f) \triangleq \frac{|\hat{G}_{xy}(f)|^2}{\hat{G}_{xx}(f)\hat{G}_{yy}(f)}$$

which measures to what degree the measured output signal is generated by linear filtering of the measured input signal.

General model including measurement noise on input and/or output measurements



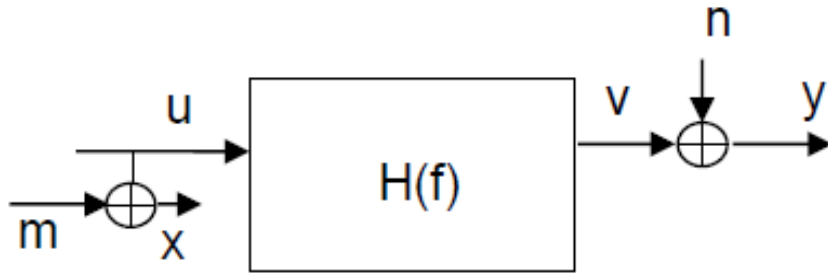
The signals are

- u : true input signal
- m : input measurement noise
- $x = u + m$: measured input signal
- v : true output signal
- n : output measurement noise
- $y = v + n$: measured output signal

Note that **m and n are additive noise on measurements (destroying the measurements)**

Often the **input signal u is also white or broadband noise to trigger all system frequencies**, but **u is a desired input noise signal**, while **measurement noises m and n are undesired**

General model with only output measurement noise (most often occurring case)



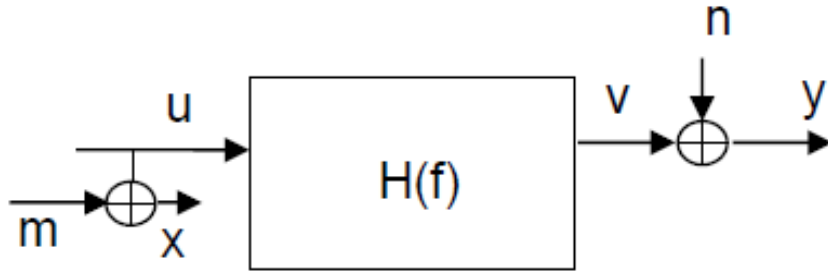
only output noise, $m = 0$

Optimal estimator is (details on blackboard) $\hat{H}(f) = \hat{H}_1(f) \stackrel{\text{def}}{=} \frac{G_{xy}(f)}{G_{xx}(f)}$

Bias $\hat{H}_1(f) = H(f) \frac{G_{uu}(f)}{G_{uu}(f) + G_{mm}(f)}$ (unbiased for $m = 0$)
(underestimates for $m \neq 0$)

Coherence function $\gamma_{xy}^2(f) = \frac{1}{1 + \frac{G_{nn}(f)}{G_{vv}(f)}} = \frac{1}{1 + \frac{1}{SNR_o(f)}}$

General model with only input measurement noise



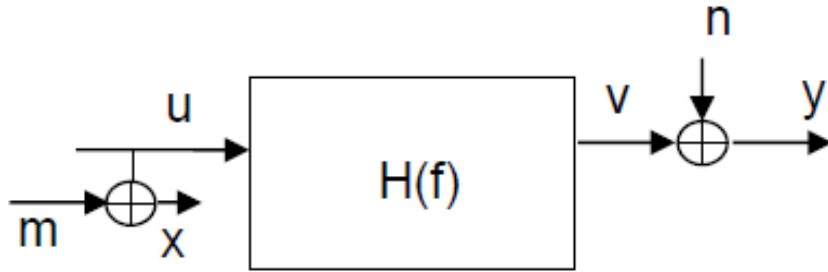
only input noise, $n = 0$

Optimal estimator is (details on blackboard) $\hat{H}(f) = \hat{H}_2(f) \stackrel{\text{def}}{=} \frac{G_{yy}(f)}{G_{xy}^*(f)}$

Bias $\hat{H}_2(f) = H(f) \frac{G_{vv}(f) + G_{nn}(f)}{G_{vv}(f)}$ (unbiased for $n = 0$)
(overestimates for $n \neq 0$)

Coherence function $\gamma_{xy}^2(f) = \frac{1}{1 + \frac{G_{mm}(f)}{G_{uu}(f)}} = \frac{1}{1 + \frac{1}{\text{SNR}_i(f)}}$

General model with both output and input measurement noise



both input and output noise

Optimal estimator (details on blackboard)

It turns out that neither $H_1(f)$ or $H_2(f)$ can be used without knowledge of either m or n

Compromise

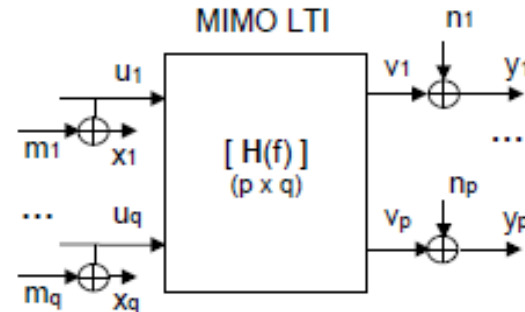
Consider either output or input noise to be dominant and use $H_1(f)$ or $H_2(f)$, respectively

Often good approximation, use $\hat{H}_{dB}(f) = \frac{1}{2} \left(\hat{H}_{1,dB}(f) + \hat{H}_{2,dB}(f) \right)$

Coherence function $\gamma_{xy}^2(f) = \left(\frac{1}{1 + \frac{G_{mm}(f)}{G_{uu}(f)}} \right) \left(\frac{1}{1 + \frac{G_{nn}(f)}{G_{vv}(f)}} \right) = \left(\frac{1}{1 + \frac{1}{SNR_i(f)}} \right) \left(\frac{1}{1 + \frac{1}{SNR_o(f)}} \right)$

Non-parametric SYSTEM identification of MIMO systems - Intro

MIMO-system with input & output measurement noise



- MIMO system outputs $v_1, v_2, \dots, v_p$ are all generated by summing of linearly filtered MIMO system inputs $u_1, u_2, \dots, u_q$
- The MIMO system is characterized by an $(p \times q)$ transfer function matrix $H(f)$
- $m_1, m_2, \dots, m_q$ are the input measurement noises
- $n_1, n_2, \dots, n_p$ are the output measurement noises
- $x_1, x_2, \dots, x_q$ are the noisy input measurements
- $y_1, y_2, \dots, y_p$ are the noisy output measurements

Case 1: only output measurement noise, $\forall m_i = 0$

$$\hat{H}_1(f) = G_{xy}(f) G_{xx}^{-1}(f)$$

$(p \times q) \quad (p \times q) \quad (q \times q)$

Case 2: only input measurement noise, $\forall n_i = 0$

$$\hat{H}_2(f) = G_{yy}(f) G_{yx}^{-1}(f)$$

$(p \times p) \quad (p \times p) \quad (p \times p)$